MIL-C-14814B(OS)
10 January 1978
SUPERSEDING
MIL-C-14814A(MU)
2 July 1973

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, M270

This specification is approved for use by the Naval Sea Systems Command (OS), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers M270 impulse cartridges for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-60539	Coating, Anodic, Conventional for Aluminum and Aluminum Alloys
MIL-C-83125	Cartridges for Cartridge Actuated/ Propellant Actuated Devices, General Design Specification for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (Code 611), Indian Head, Md. 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD, MD (Code Ident 14083)

C 8597235	Ring, Retaining
C 9234228	Carton, Alternative Packing Ammunition for Cartridge, Impulse, M270
C 9234229	Carton, Packing Ammunition, for Cartridge, Impulse, M270
C 9234230	Box, Packing, Ammunition, for Cartridge, Impulse, M270
D 10524680	Case, Cartridge
C 10533830	Disc, Sealing
D 10551427	Cartridge Head
D 10551428	Cartridge Head Assembly
F 10551431	Cartridge, Impulse, M270, Assembly
* F 11820864	Primer, Percussion

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - 1962	Surface Texture (Surface Roughness, Waviness, and Lay)

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018).

3. REQUIREMENTS

3.1 First article. When specified in the contract (see 6.2), * the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3).

3.2 Conformance to documents. Materials, parts and assemblies shall conform to drawing F 10551431 and the applicable drawings and referenced specifications.

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ions the Government reserves the right to perform any of the inspect

* 3.3 Vibration. The assembled cartridge shall be capable of withstanding the vibration test of MIL-C-83125 without evidence of disintegration of the primer, propellant, igniter or derangement of components.

* 3.4 Ballistics. The cartridge when assembled in a CKU-7/A Catapult shall propel a 459 pound mass horizontally in accordance with the requirements of TABLE I.

* TABLE I. Performance limits in CKU-7/A rocket catapult.

Temperature	-65° to -70°F	65°F to 70°F	160° to 165°F
Velocity (ft/s), max	37.5	40	42
Acceleration (g), max	20	18	20
Rate change of acceleration (g/s), max	300	300	300
Ignition delay (ms), max	30	30	30
Launch time (ms), max	210	200	180
Axial thrust (lb_f), max	7000	7000	7300

3.5 Protective finish. Protective finish shall be in accordance with MIL-C-60539.

3.6 Workmanship. The requirements of workmanship shall be as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

* 3.6.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

* 4.3.1 Initial production sample. Unless otherwise specified in the contract (see 6.2), an initial production sample consisting of 76 assembled cartridges to be used for the hardness test (4.7.1), the ballistic test (4.7.4), and sealant application tests (4.7.5), and 10 sets of unassembled components for visual and dimensional examination and other tests specified herein shall be submitted to an activity designated by the procuring activity (see 6.2) for initial production examination and test.

4.3.1.1 Examination and tests. After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications.

4.3.1.2 Initial production sample failure. Failure of the sample to comply with the requirement of the drawings and specifications shall result in initial production sample disapproval.

* 4.4 Quality conformance inspection. The quality conformance inspection shall consist of the examinations of 4.5 and the tests of 4.7 performed on samples selected in accordance with 4.4.1 and 4.4.2.

4.4.1 Sampling. Unless otherwise specified, sampling shall be in accordance with the provisions of MIL-STD-105 and 4.6.

4.4.2 Size of lot. An ammunition lot shall consist of an identifiable group of metal part assemblies consisting of:

a. Components produced by a homogeneous process.
b. Propellant from not more than one lot of ingredients.
c. Primers from not more than one lot.
d. Igniters from not more than one lot.

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4.5 Quality conformance examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in TABLES II and III. For an individual defect (single characteristic), the criteria in TABLE II shall be used. For total defects within a major or minor class, the criteria in TABLE III shall be used. The defects shall be classified in accordance with TABLE IV. All non-conforming material shall be rejected. The roughness comparison specimens prescribed in ANSI B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II. Sampling plan (single characteristic).

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1-300	100% Examination	-	-	-	-
301-1300	110	1	2	2	3
1301-3200	150	2	3	3	4
3201-8000	225	3	4	4	5
8001-22000	300	4	5	5	6

TABLE III. Total defect criteria.

Number of defects in class	Sample size			
	110	150	225	300
	Major defects allowed			
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over - 43	6	8	11	14
	Minor defects allowed			
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over - 48	8	11	17	20

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TABLE IV. Classification of defects.

Classification	Defect	Method of Inspection
* Primer (Drawing 11820864)		
CRITICAL		
1	Metal defective	Visual
2	Primer foil discolored	Visual
3	Mixed primers	Visual
4	Improper assembly	Visual
5	Mixed primers	Visual
MAJOR		
101	Burr	Visual
102	Foreign matter	Visual
MINOR		
None defined		
Ring, Retaining (Drawing C 8597235)		
CRITICAL		
None defined		
MAJOR		
101	Concentricity of inside diameter with outside diameter	Gage
102	Diameter, outside, min	Gage
103	Diameter, inside, max	Gage
104	Metal defective	Visual
105	Protective finish inadequate or defective	Visual
MINOR		
201	Diameter, outside, max	Gage
202	Diameter, inside, min	Gage
203	Thickness	Gage
204	Surface finish	Visual

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TABLE IV. Classification of defects. - continued

Classification	Defect	Method of Inspection
	Case, Cartridge (Drawing D 10524680)	
CRITICAL		
None defined		
MAJOR		
101	Concentricity of head with body	Gage
102	Straightness of base	Gage
103	Diameter of body, min	Gage
104	Diameter of head, min	Gage
105	Thickness of base	Gage
106	Thickness of sidewall at head	Gage
107	Protective finish inadequate or defective	Visual
108	Burr or sharp corner	Visual
109	Foreign matter	Visual
MINOR		
201	Diameter of body, max	Gage
202	Diameter of head, max	Gage
203	Location of datum diameter	Gage
204	Radius or chamber missing or incorrect	Visual
205	Surface finish improper	Visual
	Disc, Sealing (Drawing C 10533830)	
CRITICAL		
None defined		
MAJOR		
101	Diameter, min	Gage
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr	Visual
105	Foreign matter	Visual
MINOR		
201	Diameter, max	Gage
202	Thickness	Gage

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TABLE IV. Classification of defects. - continued

Classification	Defect	Method of Inspection
Cartridge Head (Drawing D 10551427)		
CRITICAL		
1	Metal defective	Visual
MAJOR		
101	Concentricity of head with primer pocket	Gage
102	Concentricity of head with body	Gage
103	Concentricity of cartridge head protrusion with primer pocket	Gage
104	Concentricity of primer pocket recess with primer pocket	Gage
105	Diameter of primer pocket	Gage
106	Diameter of primer pocket recess	Gage
107	Depth of primer pocket	Gage
108	Thickness of wall at open end of counterbore	Gage
109	Thickness of wall at open end of cavity	Gage
110	Thickness of wall at base of primer pocket recess	Gage
111	Thickness of head	Gage
112	Location of datum diameter	Gage
113	Surface finish of primer pocket improper	Visual
114	Protective finish inadequate or defective	Visual
115	Burr, except where permitted	Visual
116	Foreign matter	Visual
MINOR		
201	Diameter of head	Gage
202	Diameter cartridge head protrusion	Gage
203	Diameter of body	Gage
204	Depth of counterbore	Gage
205	Height of cartridge head protrusion	Gage
206	Total length	Gage
207	Radius or chamfer missing or incorrect	Visual
208	Surface finish improper, except as otherwise classified	Visual

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TABLE IV. Classification of defects. - continued

Classification	Defect	Method of Inspection
Cartridge Head Assembly (Drawing D 10551428)		
CRITICAL		
1	Any part missing	Visual
2	Improper assembly	Visual
3	Metal defective	Visual
4	Primer not fully seated	Gage
5	Primer not below flush with surfaces	Gage
MAJOR		
101	Diameter of body, min	Gage
102	Junction top edge of primer and pocket not properly sealed and crimped	Visual
MINOR		
201	Diameter of body, max	Gage
202	Total length, max	Gage
Carton, Packing Ammunition, for Cartridge, Impulse, M270 (Drawing C 9234229), or Carton, Alternative Packing Ammunition, for Cartridge, Impulse, M270 (Drawing C 9234228)		
CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201	Contents loose	Manual/Visual
202	Quantity of contents incorrect	Visual
203	Separator or filler missing	Visual
204	Head sealer incomplete, loose or distorted	Visual
205	Cut or damaged through impregnated layers	Visual
206	Marking missing, incorrect or illegible	Visual

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TABLE IV. Classification of defects. - continued

Classification	Defect	Method of Inspection
	Box, Packing, Ammunition, for Cartridge, Impulse, M270 (Drawing C 9234230)	
CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201	Contents loose	Manual/Visual
202	Hardware missing, loose, broken or improperly engaged	Visual
203	Strapping missing, loose, broken or improperly assembled	Visual
204	Content of box exposed	Visual
205	Box damaged	Visual
206	Marking missing, incorrect or illegible	Visual
207	Number of cartons incorrect	Visual
208	Car seal missing, unsealed or improperly positioned	Visual

4.6 Sampling for tests. Sampling for tests shall be in accordance with the following:

4.6.1 Hardness. Fifty units from each heat treated lot shall be selected for this hardness test. Failure of any unit to comply with drawing requirements shall be cause for rejection of the lot.

4.6.2 Protective finish. Unless otherwise specified, sampling plans and procedures used in determination of the acceptability of coated parts and articles submitted by a supplier shall be in accordance with provisions set forth in MIL-STD-105 and MIL-C-60539.

4.6.3 Vibration. Twelve units shall be selected from the first article ballistic sample and tested in accordance with 4.7.3. Failure of any unit to comply with the requirements of 3.3 shall be cause for rejection of the lot.

4.6.4 Ballistics. The twenty-one assembled cartridges from each production lot shall be grouped and temperature conditioned in accordance with 4.7.4.1. Failure of any cartridge to comply with the requirements of 3.4 shall be cause for rejection of the lot.

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4.6.5 Sealant application. Five assembled units from each production lot shall be selected for this test. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.

4.7 Test methods and procedures.

4.7.1 Hardness. Prior to the application of the protective finish, the hardness testing machine shall be used to determine compliance with drawing requirements.

4.7.2 Protective finish. Test procedures for protective finish shall be in accordance with MIL-C-60539. The unit shall be sectioned longitudinally to better expose the critical surface to the salt spray.

* 4.7.3 Vibration. The method and procedure for performing the vibration test shall be in accordance with the vibration test of MIL-C-83125.

4.7.4 Ballistics. The ballistics test shall be performed by the Government (see 6.2). The cartridges shall be grouped and temperature conditioned in accordance with 4.7.4.1 and TABLE V and rigidly supported in the test fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time is exceeded, the sample shall be reconditioned in accordance with 4.7.4.1.

4.7.4.1 Temperature conditioning. A standard inert M270 cartridge containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

TABLE V. Temperature conditioning sample size.

Temperature	Sample Size
-65° to -70°F	7
65° to 70°F	7
160° to 165°F	7

4.7.5 Sealant application. The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of the joining surfaces at the junction of the sealing disc and case sidewall and that the propellant cavity is free of sealant.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be Level A or Level B as specified by the procuring agency (see 6.2).

5.1.1 Level A. Preservation, packaging, packing and marking shall be in accordance with Drawings C 9234229 and C 9234230.

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5.1.2 Level B. Preservation, packaging, packing and marking shall be in accordance with Drawings C 9234229 and C 9234228.

6. NOTES

6.1 Intended use. The M270 impulse cartridge is intended to be used on the CKU-7/A aircraft ejection seat catapult.

6.2 Ordering data. Procurement documents should specify the following:

* 6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Whether a first article sample is required (see 3.1).
c. Quantity required (including first article samples).
d. Assigned activity for first article inspection.
e. Government activity assigned for ballistics test.
f. The level of packaging required.

* 6.2.2 Contract data requirements. Any data required for delivery in connection with this document shall be specified on a DD Form 1423 incorporated into the contract. Such data will be delivered as identified on completed (numbered) DIDs (Data Item Descriptions/DD Form 1664) which will be documented in the applicable ADL (Authorized Data List).

6.3 Delay time. Delay time is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

6.4 Hazard notice. The cartridge described herein and certain of the components are flammable and/or explosive, consequently presenting hazards in manufacture, handling, storage and shipment. The contractor should recognize explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

Loaded Assembly Hazard Information:
Explosive Hazard Classification: Class 3
Storage Compatibility: Group B
Department of Transportation (DOT) Hazard Class
DOT Marking - Explosive Power Devices - Class
Handle Carefully - (Keep Fire Away)

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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